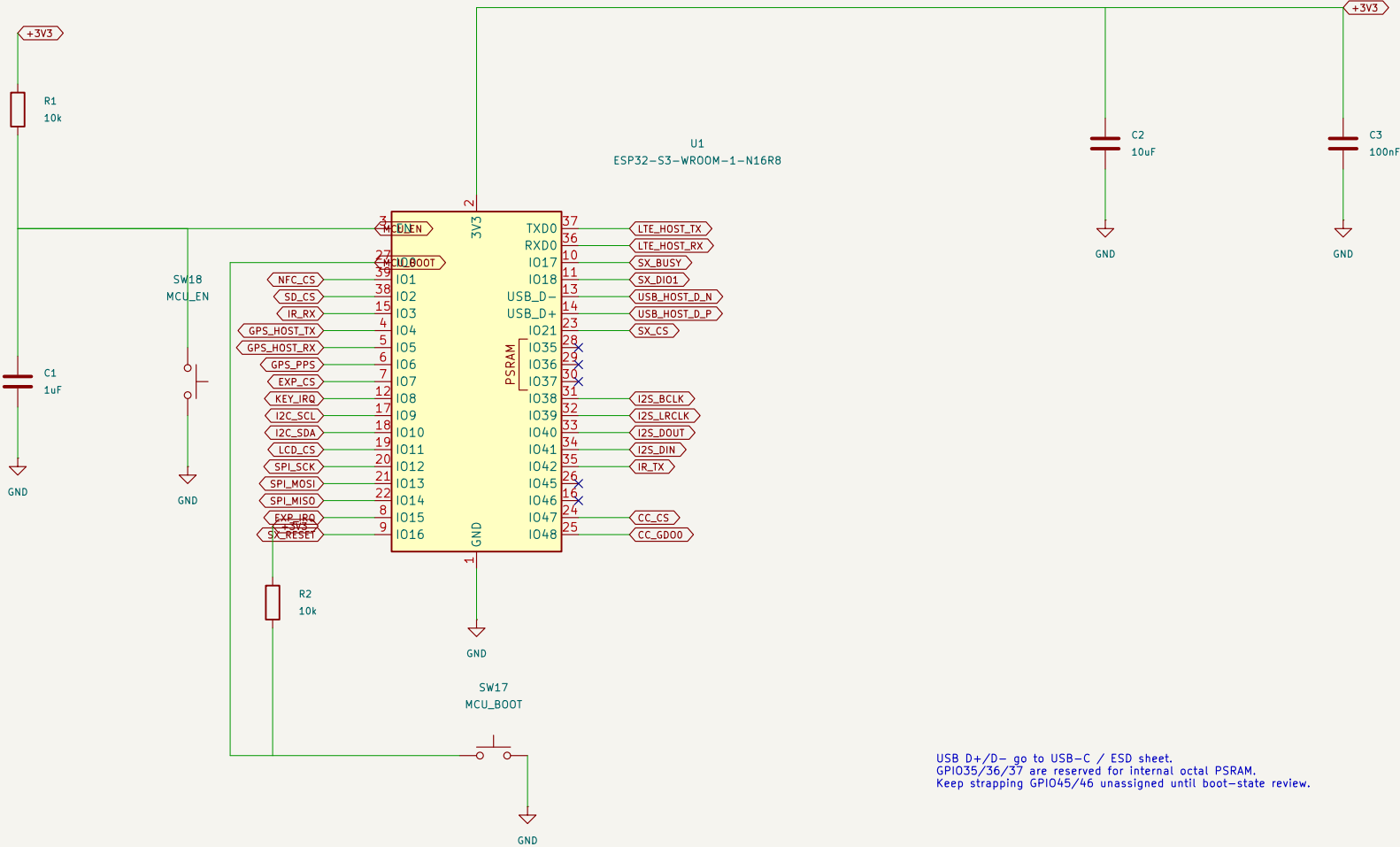


| HANDSET REV A / SYSTEM SCHEMATIC INDEX                                                                     |                                           |  |  |                                                |  |  |                                              |  |  |                                                                                                                                                  |  |
|------------------------------------------------------------------------------------------------------------|-------------------------------------------|--|--|------------------------------------------------|--|--|----------------------------------------------|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Engineering draft. Retained ESP32-S3 / MakerFocus 3000 mAh / external display / all peripheral interfaces. |                                           |  |  |                                                |  |  |                                              |  |  |                                                                                                                                                  |  |
| A                                                                                                          | ESP32 host, boot and reset                |  |  | Key matrix and trackball                       |  |  | USB-C data and ESD                           |  |  |                                                                                                                                                  |  |
|                                                                                                            | File: core.kicad_sch                      |  |  | File: keys.kicad_sch                           |  |  | File: usb.kicad_sch                          |  |  |                                                                                                                                                  |  |
| B                                                                                                          | Fused USB Input and efuse                 |  |  | USB-C source detection and permission          |  |  | Battery, thermistor and programmable charger |  |  |                                                                                                                                                  |  |
|                                                                                                            | File: usb-input.kicad_sch                 |  |  | File: usb-detect.kicad_sch                     |  |  | File: power.kicad_sch                        |  |  |                                                                                                                                                  |  |
| C                                                                                                          | Power control and fuel gauge              |  |  | Main 3.3 V buck-boost supply                   |  |  | Auxiliary 5 V and 1.8 V supplies             |  |  |                                                                                                                                                  |  |
|                                                                                                            | File: power-control.kicad_sch             |  |  | File: reg3.kicad_sch                           |  |  | File: aux-power.kicad_sch                    |  |  |                                                                                                                                                  |  |
| D                                                                                                          | Regulated modem supply and enable         |  |  | SX1262 LoRa radio                              |  |  | CC1101 sub-GHz radio                         |  |  |                                                                                                                                                  |  |
|                                                                                                            | File: modem-power.kicad_sch               |  |  | File: radios.kicad_sch                         |  |  | File: cc-radio.kicad_sch                     |  |  |                                                                                                                                                  |  |
| E                                                                                                          | 125 kHz reader analog circuit             |  |  | LF reader logic translation                    |  |  | GNSS receiver and passive antenna            |  |  |                                                                                                                                                  |  |
|                                                                                                            | File: lf-radio.kicad_sch                  |  |  | File: lf-support.kicad_sch                     |  |  | File: gnss-antennas.kicad_sch                |  |  |                                                                                                                                                  |  |
| F                                                                                                          | NFC reader, oscillator and local supplies |  |  | Media speaker amplifier and digital microphone |  |  | Cellular adapter and call-audio harnesses    |  |  |                                                                                                                                                  |  |
|                                                                                                            | File: nfc.kicad_sch                       |  |  | File: audio.kicad_sch                          |  |  | File: interfaces.kicad_sch                   |  |  |                                                                                                                                                  |  |
| G                                                                                                          | Protected external module port            |  |  | MicroSD and case-mounted infrared              |  |  | External display, touch and backlight        |  |  |                                                                                                                                                  |  |
|                                                                                                            | File: protection.kicad_sch                |  |  | File: storage-ir.kicad_sch                     |  |  | File: display.kicad_sch                      |  |  |                                                                                                                                                  |  |
| H                                                                                                          | Always-on charging supervisor             |  |  | Display supply and translation                 |  |  | Expansion control and isolation              |  |  | <div>Sheet: /<br/>File: handset.kicad_sch</div> <div><div>Title:</div><div>Size: A2   Date:</div><div>KiCad E.D.A. 10.0.6   Id: 1/25</div></div> |  |
|                                                                                                            | File: power-supervisor.kicad_sch          |  |  | File: display-interface.kicad_sch              |  |  | File: expansion-control.kicad_sch            |  |  |                                                                                                                                                  |  |

ESP32 host, boot and reset

Retain N16R8. Module pads 28/29/30 belong to octal PSRAM. USB programming uses the USB sheet.



USB D+/D- go to USB-C / ESD sheet.  
GPIO35/36/37 are reserved for internal octal PSRAM.  
Keep strapping GPIO45/46 unassigned until boot-state review.

RESET / EN: 10k pull-up + 1uF  
BOOT held low during reset selects download mode.

Circuit review in progress – not released for fabrication

Sheet: /ESP32 host, boot and reset/  
File: core.kicad\_sch

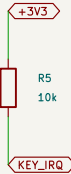
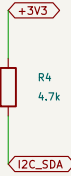
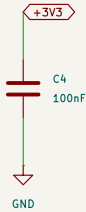
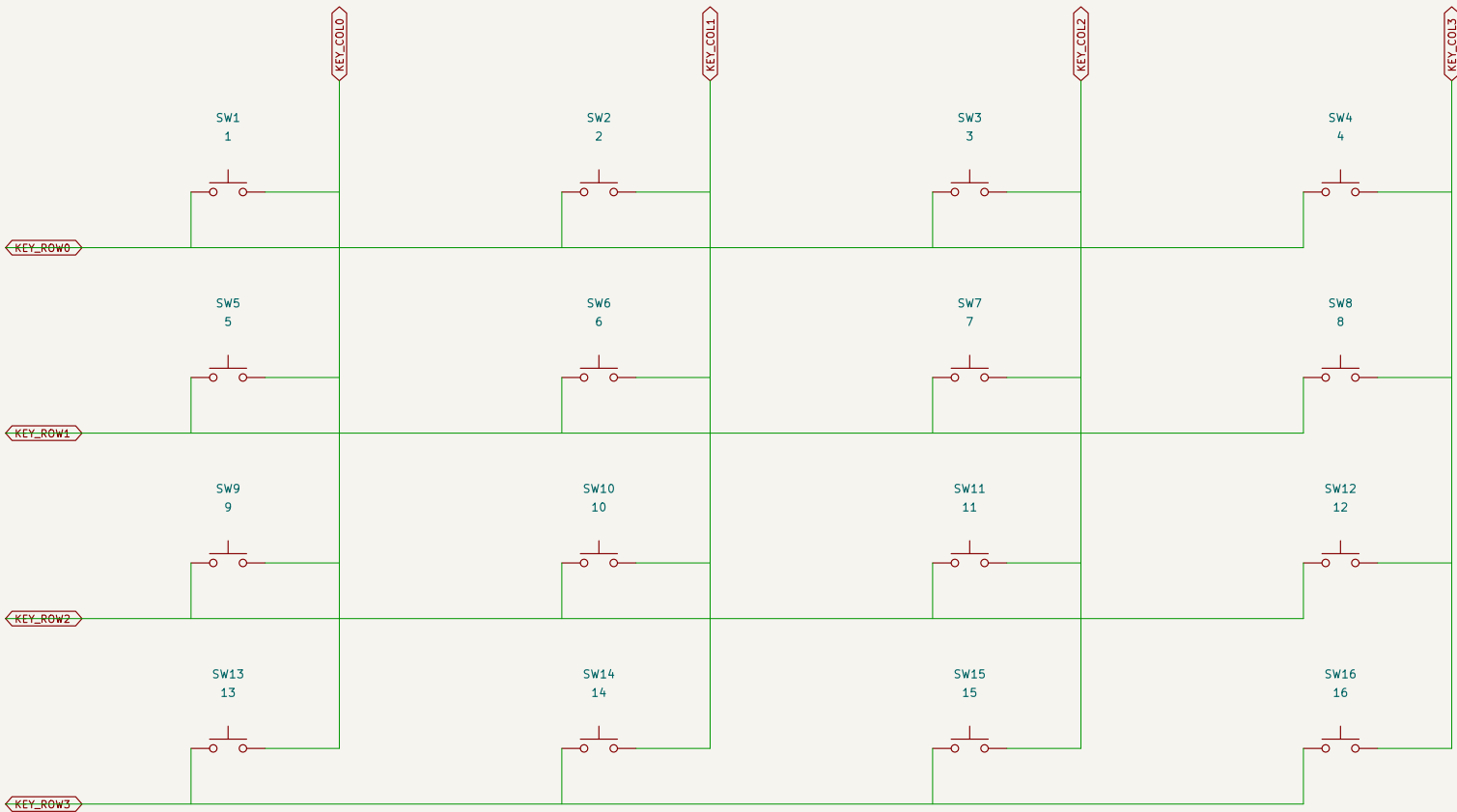
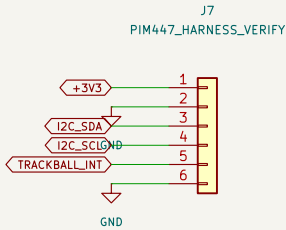
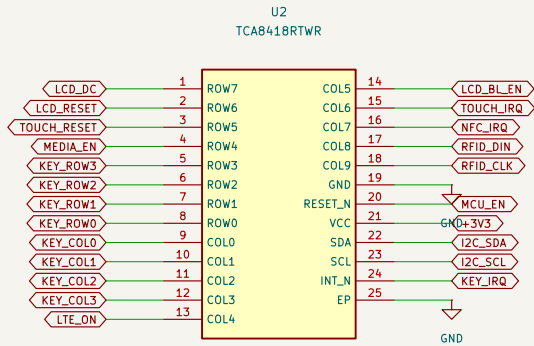
**Title: ESP32 host, boot and reset**

Size: A3  
Date: 2026-09-24

Rev: A – engineering draft  
Id: 2/25

Key matrix and trackball

TCAB418: 4 rows x 4 columns; remaining pins are firmware-configured GPIO. Confirm PIM447 harness orientation.



Circuit review in progress – not released for fabrication

Sheet: /Key matrix and trackball/  
File: keys.kicad\_sch

**Title: Key matrix and trackball**

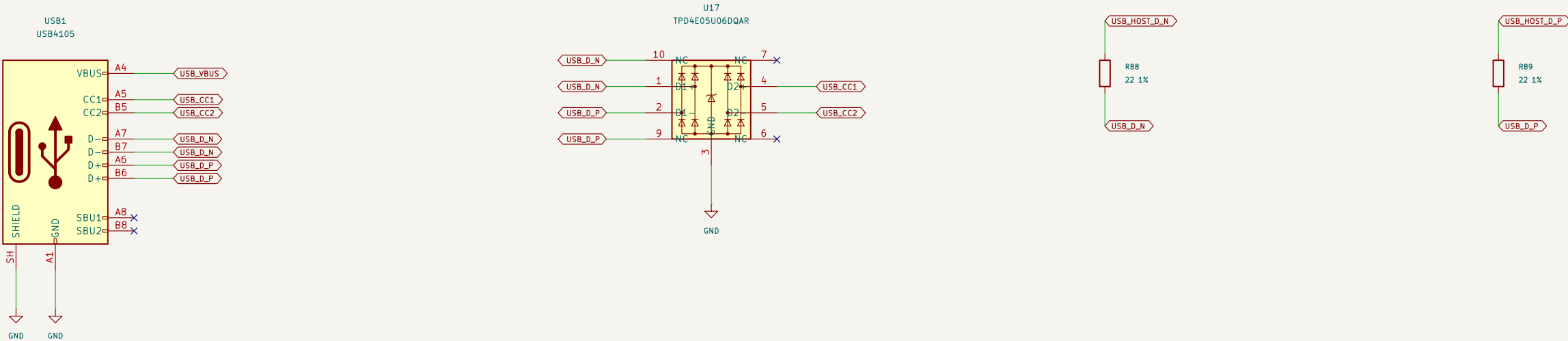
Size: A3  
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft  
Id: 3/25

USB-C data and ESD

U17 NC lands 10/9 externally bridge to 1/2 for pass-through routing; no internal connection. R88/R89 host series termination, initial 22 ohms; qualify on hardware.



Circuit review in progress – not released for fabrication

Sheet: /USB-C data and ESD/  
File: usb.kicad\_sch

**Title: USB-C data and ESD**

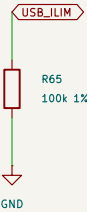
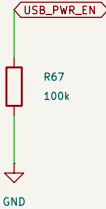
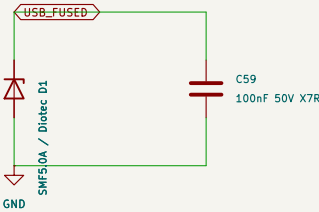
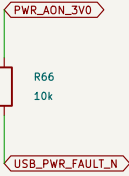
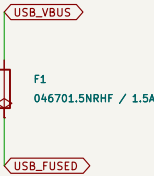
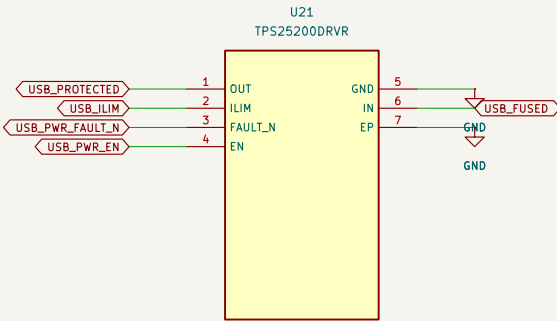
Size: A3  
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft  
Id: 4/25

Fused USB input and eFuse

TPS25200 R\_ILIM = 100k, 1%. F1/TVS pulse coordination requires bench validation. No USB PD negotiation.



Supply flag: external USB VBUS through F1. No internal source is implied.

Circuit review in progress – not released for fabrication

Sheet: /Fused USB input and eFuse/  
File: usb-input.kicad\_sch

**Title: Fused USB input and eFuse**

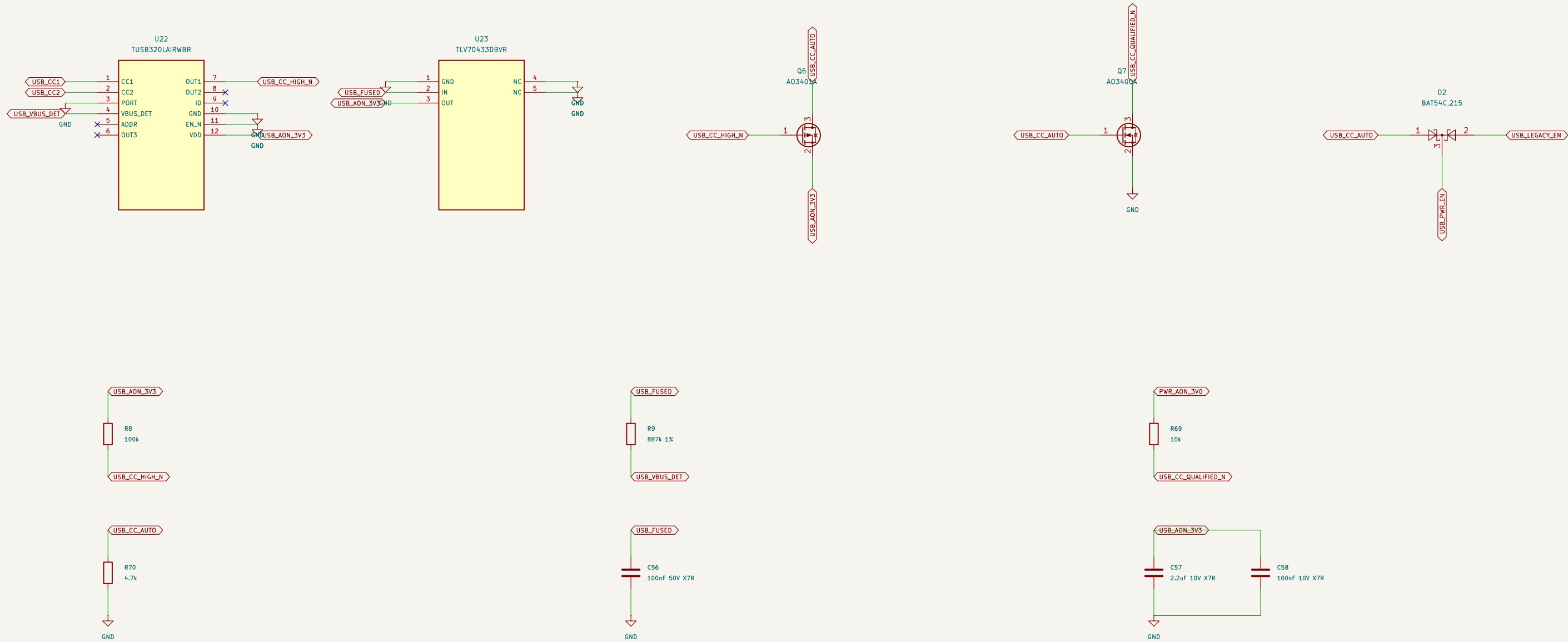
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KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft  
Id: 5/25

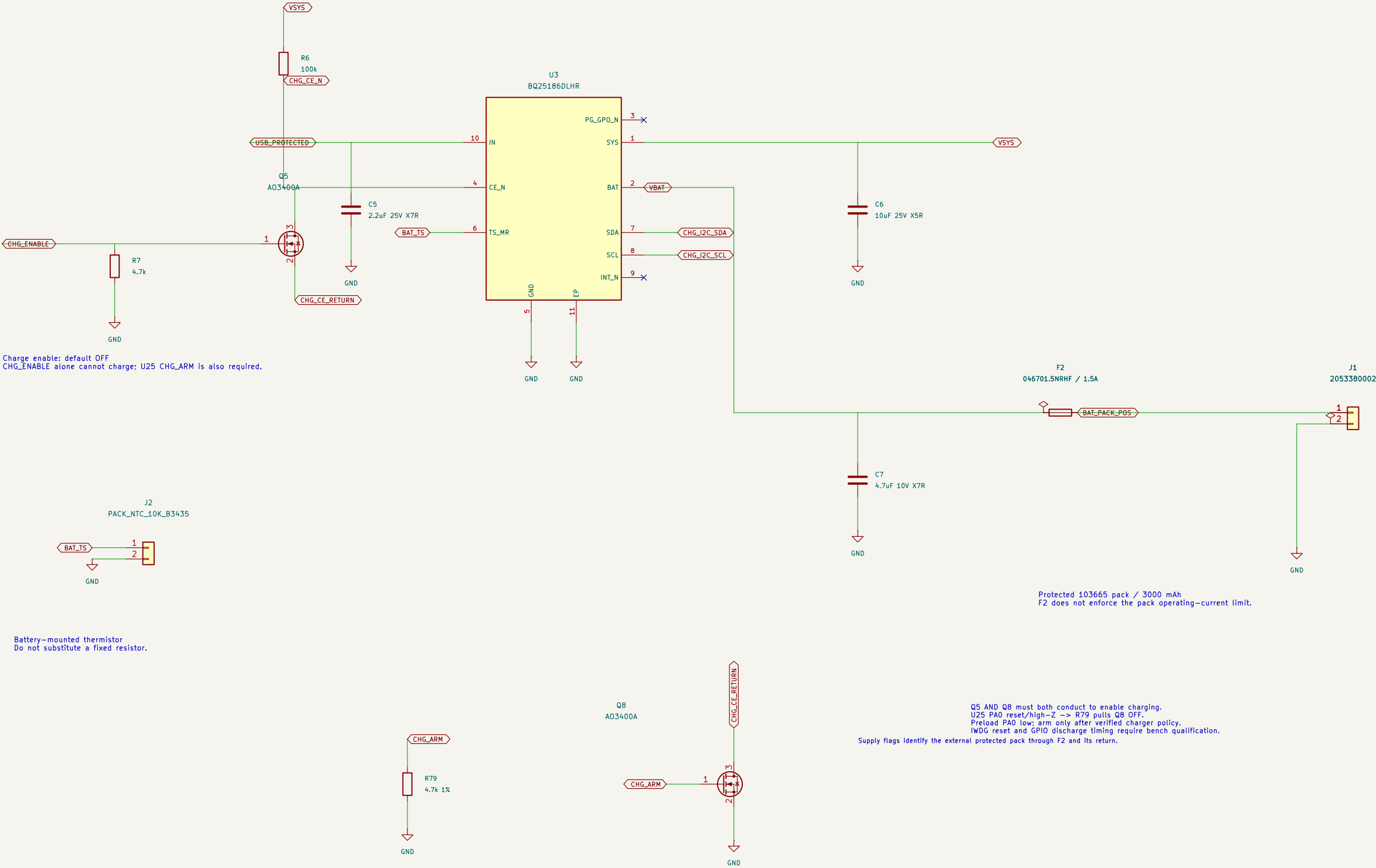
USB-C source detection and permission

Sink-only internal Rd. OUT1 low authorizes Type-C 1.5A/3A. Legacy permission requires USB enumeration; see charger-policy.json.



Battery, thermistor and programmable charger

Verify actual battery polarity. J2 is a battery-mounted 10k B3435 NTC. CE defaults OFF. Firmware must apply charger-policy.json.



Circuit review in progress – not released for fabrication

Sheet: /Battery, thermistor and programmable charger/  
File: power.kicad\_sch

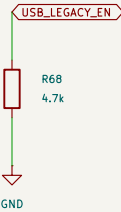
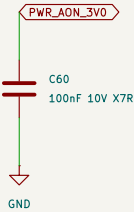
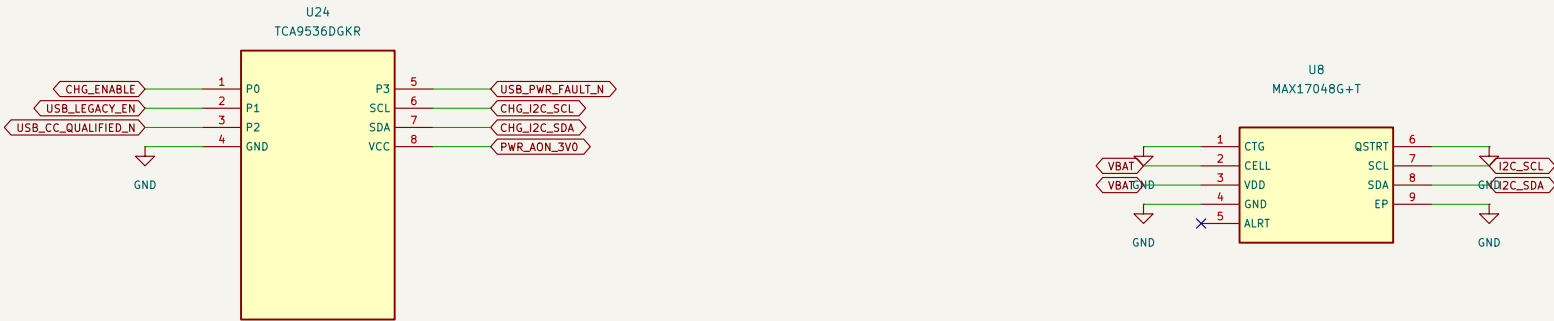
**Title: Battery, thermistor and programmable charger**

Size: A3  
Date: 2026-09-24

Rev: A – engineering draft  
Id: 7/25

Power control and fuel gauge

I2C: TCA9536 0x41, MAX17048 0x36, charger 0x6A. Preload low output latches before enabling charger—control outputs.



Circuit review in progress – not released for fabrication

Sheet: /Power control and fuel gauge/  
File: power-control.kicad\_sch

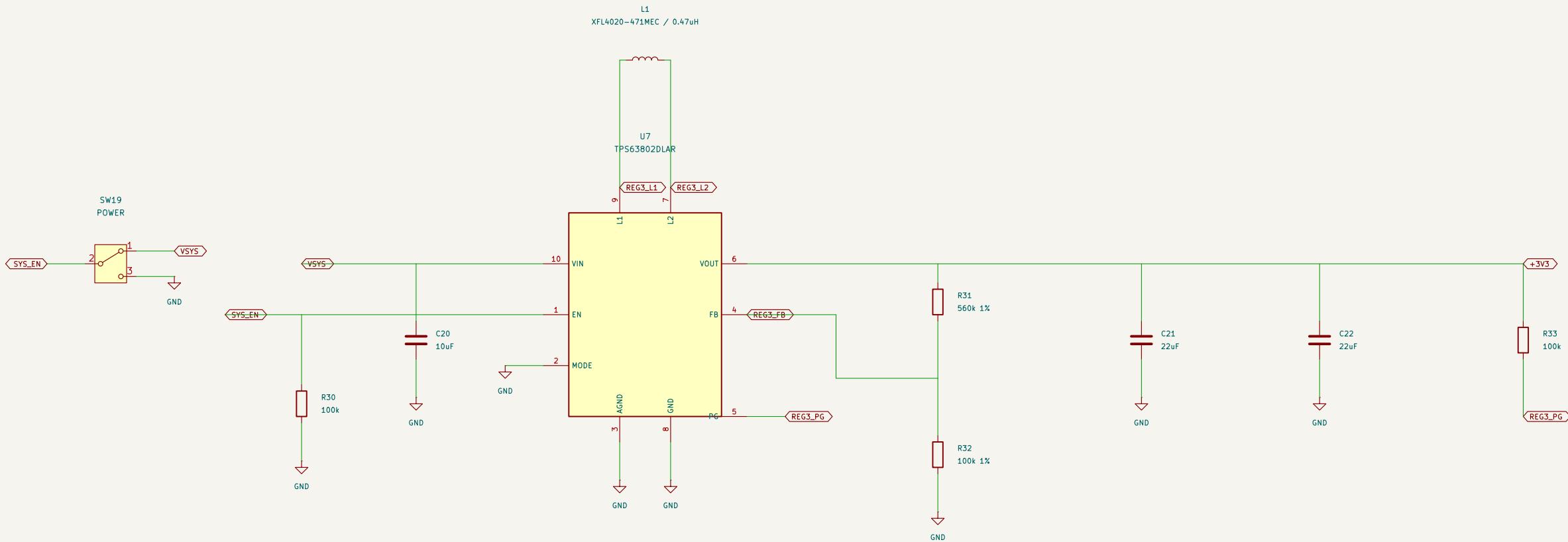
**Title: Power control and fuel gauge**

Size: A3 Date: 2026-09-24  
KiCad E.D.A. 10.0.6

Rev: A – engineering draft  
Id: 8/25

Main 3.3 V buck–boost supply

TPS63802 local PCB copper is routed. External distribution, effective capacitance, thermal limits and load transients remain to qualify.



0.6 V reference x (1 + 560k / 100k) = 3.3 V nominal.  
Feedback senses the output capacitor node; keep it clear of the switching loop.

SW19 controls SYS\_EN.  
Battery / charger circuitry can remain energized while logic rails are off.

Circuit review in progress – not released for fabrication

Sheet: /Main 3.3 V buck–boost supply/  
File: reg3.kicad\_sch

**Title: Main 3.3 V buck–boost supply**

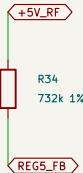
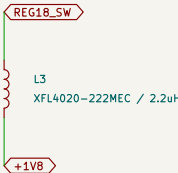
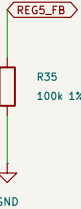
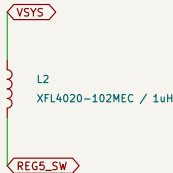
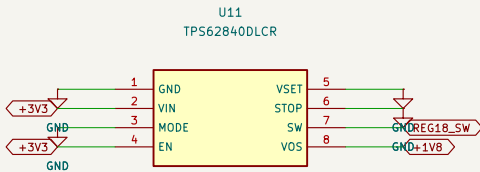
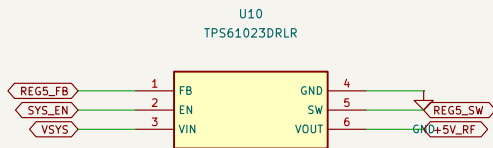
Size: A3  
KiCad E.D.A. 10.0.6

Date: 2026–09–24

Rev: A – engineering draft  
Id: 9/25

Auxiliary 5 V and 1.8 V supplies

Switching loops must be short in PCB layout. The 5 V rail also supplies display backlight and the LF reader.



Circuit review in progress – not released for fabrication

Sheet: /Auxiliary 5 V and 1.8 V supplies/  
File: aux-power.kicad\_sch

**Title: Auxiliary 5 V and 1.8 V supplies**

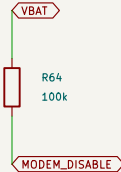
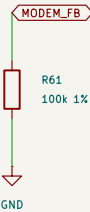
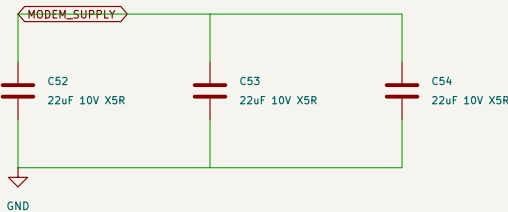
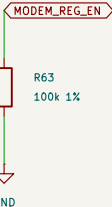
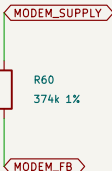
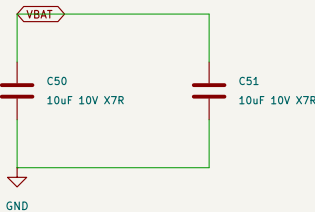
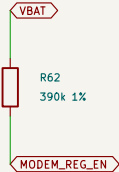
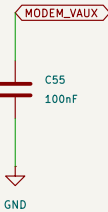
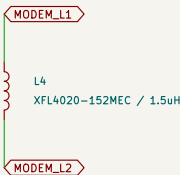
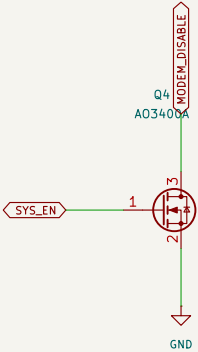
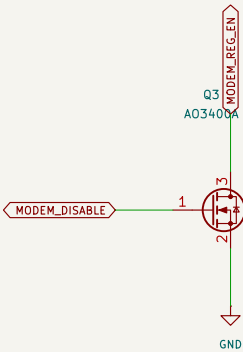
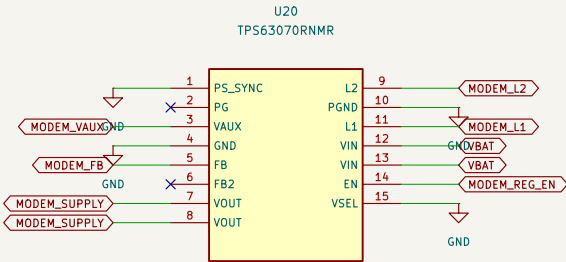
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KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft  
Id: 10/25

Regulated modem supply and enable

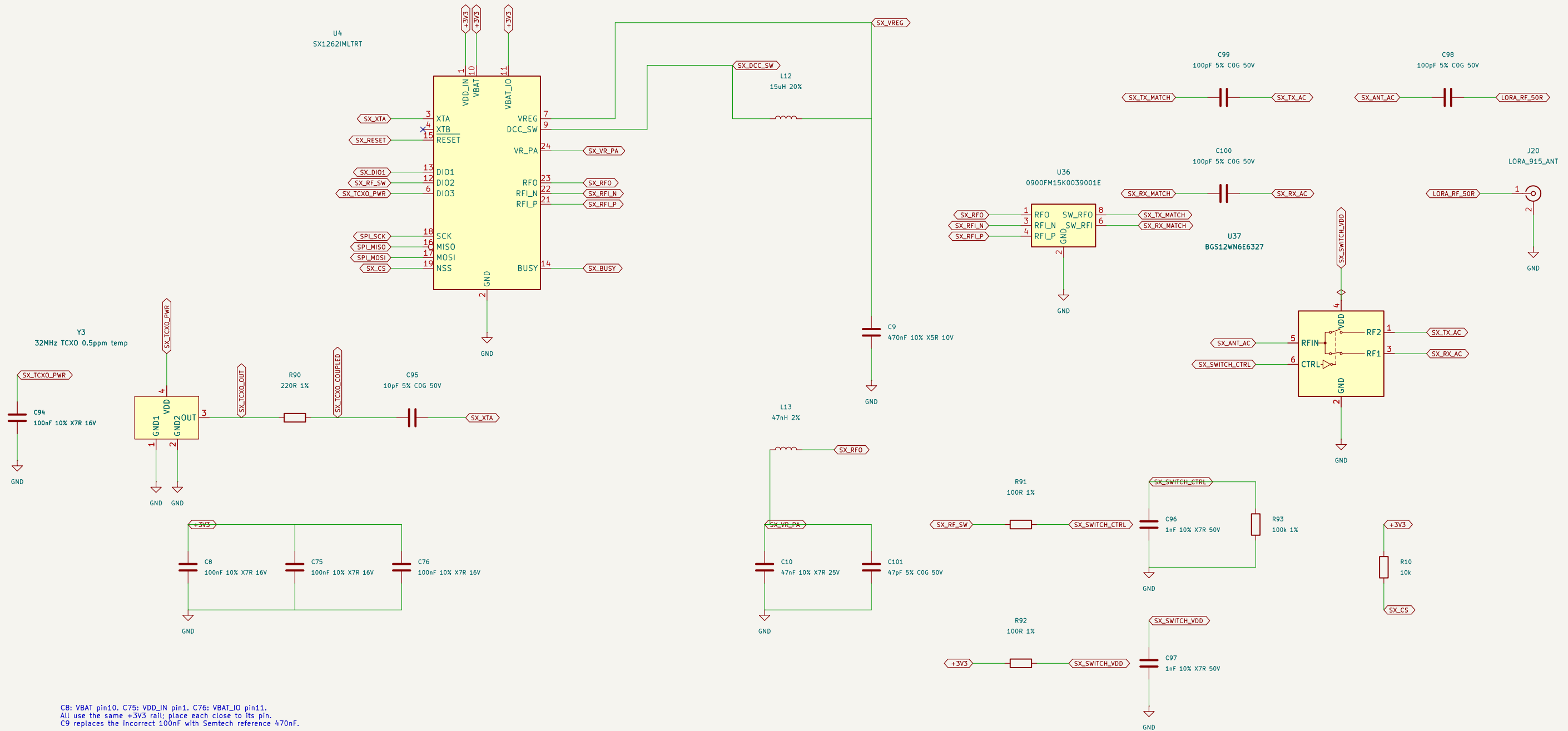
Nominal output 3.792 V. Exact modem burst current and MakerFocus pack current budget are unresolved release blockers.



|                                                                           |                  |                            |
|---------------------------------------------------------------------------|------------------|----------------------------|
| Circuit review in progress – not released for fabrication                 |                  |                            |
| Sheet: /Regulated modem supply and enable/<br>File: modem-power.kicad_sch |                  |                            |
| Title: Regulated modem supply and enable                                  |                  |                            |
| Size: A3                                                                  | Date: 2026-09-24 | Rev: A – engineering draft |
| KiCad E.D.A. 10.0.6                                                       |                  | Id: 11/25                  |

## SX1262 LoRa radio

SX1262 clock, supply, PA feed and RF matching/switch captured. RF layout incomplete; do not power.



C8: VBAT pin10. C75: VDD\_IN pin1. C76: VBAT\_I/O pin11.  
All use the same +3V3 rail; place each close to its pin.  
C9 replaces the incorrect 100nF with Semtech reference 470nF.

Y3 pads 1/2 = GND, 3 = OUT, 4 = VDD. XTB is intentionally NC.  
Configure DI03 for 1.8V; provisional startup allowance 5ms.  
Confirm TCXO output  $\leq 1.2\text{Vpp}$ ; no qualified maximum in series data.

U37: CTRL=0 selects RF1/RX; CTRL=1 selects RF2/TX.  
R93 defaults to RX. C98/C99/C100 explicitly block DC.  
100pF RF blocks are starting values, subject to RF measurement.  
No RF route or thermal/ground-via qualification in this checkpoint.

SetRegulatorMode(0x01) only in STDBY\_RC, after hardware qualification.  
L12 supplies the core, not the SX1262 high-power PA.  
L13: 47nH PA choke; C10/C101: 47nF/47pF bypass.  
RAK4270 values; final RF layout and qualification remain open.

Circuit review in progress – not released for fabrication

Sheet: /SX1262 LoRa radio/  
File: radios.kicad\_sch

**Title:** SX1262 LoRa radio

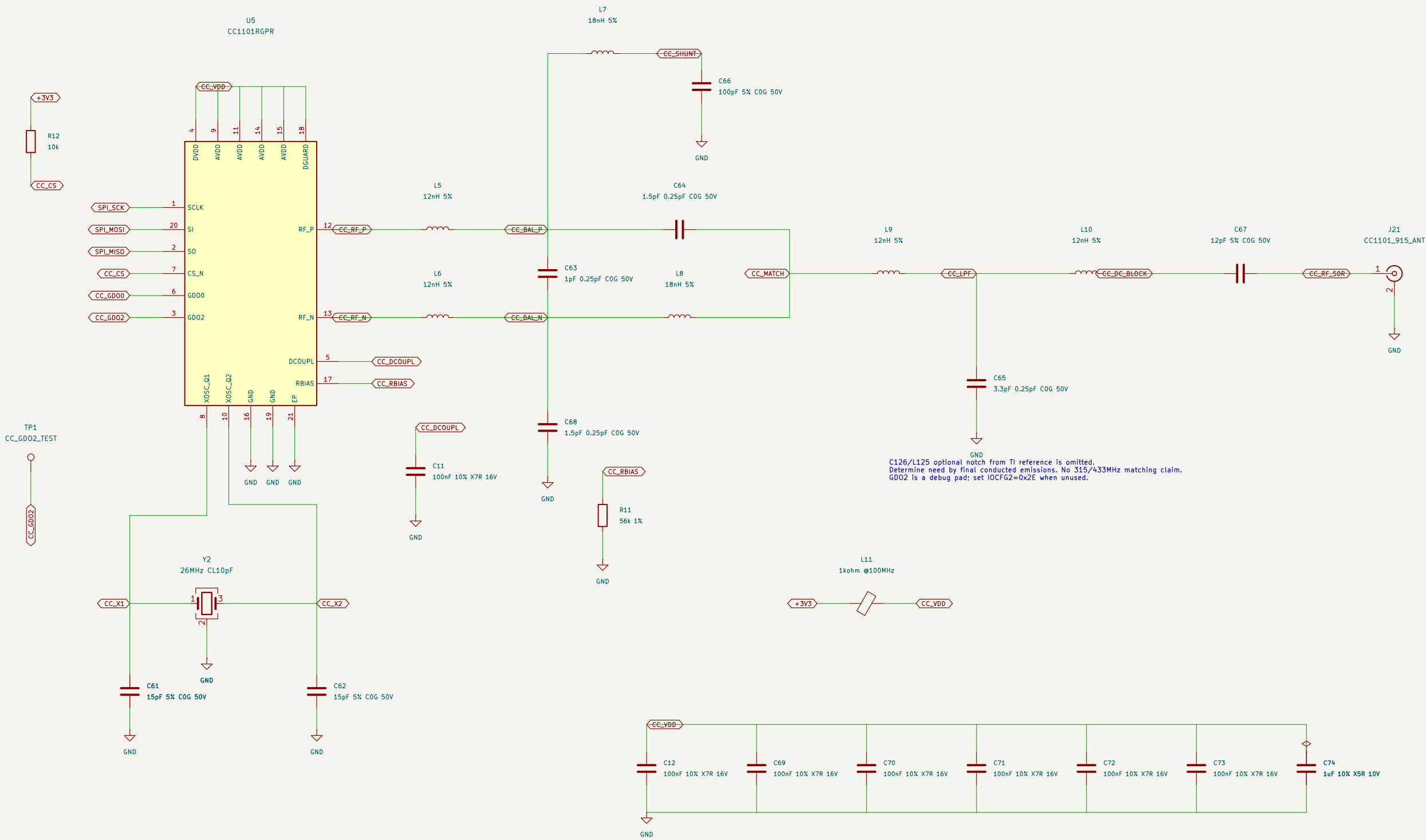
|          |                  |
|----------|------------------|
| Size: A3 | Date: 2026-10-08 |
|----------|------------------|

Rev: A – engineering draft

Id: 12/25

CC1101 sub-GHz radio

868/915MHz frontend: TI SWRR045 rev3.0 / SWRS061l Fig11. RF layout and tuning are NOT qualified. Separate 50-ohm antenna.



Y2: ABM8-26.000MHZ-10-D-1-G-T  
CL = 15pF / 2 + estimated 2.5pF stray = 10pF.  
Measure oscillator frequency, startup and drive level.

C126/L125 optional notch from TI reference is omitted.  
Determine need by final conducted emissions. No 315/433MHz matching claim.  
GD02 is a debug pad; set IOCFG2=0x2E when unused.

CC\_VDD flag: main +3V3 through passive L11; not an independent source.  
Place one 100nF at each supply pin group, plus local 1uF bulk.

Circuit review in progress – not released for fabrication

Sheet: /CC1101 sub-GHz radio/  
File: cc-radio.kicad\_sch

**Title: CC1101 sub-GHz radio**

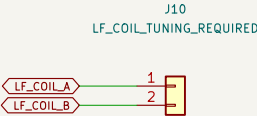
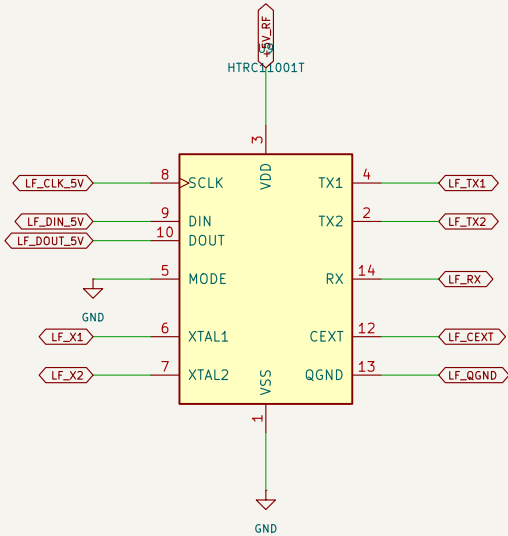
Size: A3  
Date: 2026-09-24

KiCad E.D.A. 10.0.6

Rev: A – engineering draft  
Id: 13/25

125 kHz reader analog circuit

HTRC110 requires its oscillator, QGND/CEXT bypass, resonant coil network and a timing-capable host interface.



Circuit review in progress – not released for fabrication

Sheet: /125 kHz reader analog circuit/  
File: lf-radio.kicad\_sch

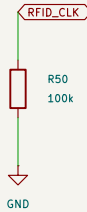
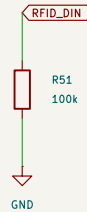
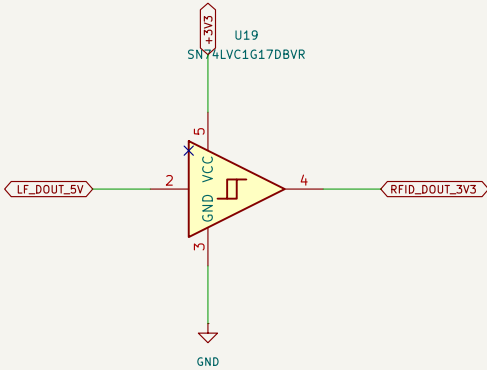
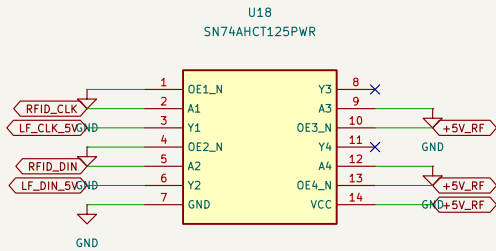
**Title: 125 kHz reader analog circuit**

Size: A3 Date: 2026-09-24  
KiCad E.D.A. 10.0.6

Rev: A – engineering draft  
Id: 14/25

LF reader logic translation

AHCT inputs accept 3.3 V logic at 5 V supply; LVC return buffer is powered at 3.3 V. Host timing remains a design requirement.



Circuit review in progress – not released for fabrication

Sheet: /LF reader logic translation/  
File: lf-support.kicad\_sch

**Title: LF reader logic translation**

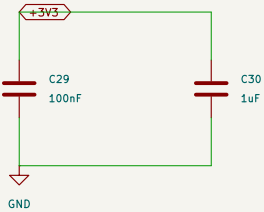
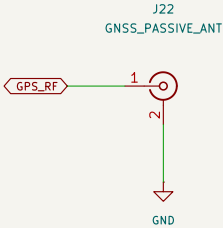
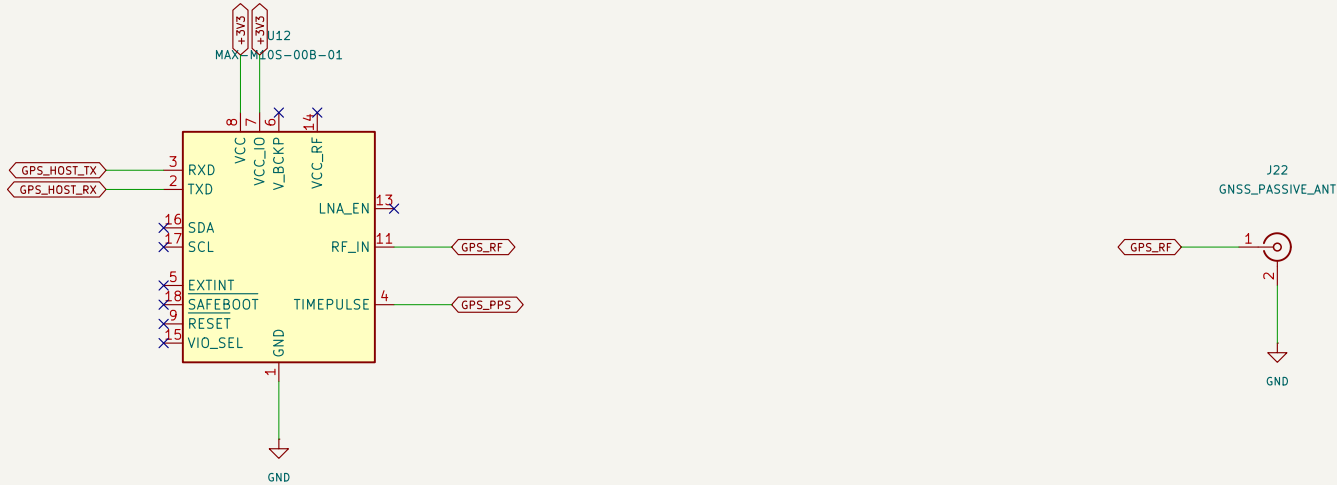
Size: A3  
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft  
Id: 15/25

GNSS receiver and passive antenna

MAX-M10S passive antenna. V\_BCKP remains open: cold starts after power removal. No direct LiPo connection.



Circuit review in progress – not released for fabrication

Sheet: /GNSS receiver and passive antenna/  
File: gnss-antennas.kicad\_sch

**Title: GNSS receiver and passive antenna**

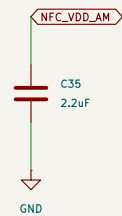
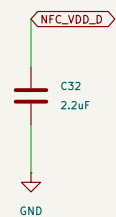
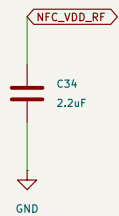
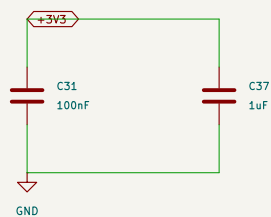
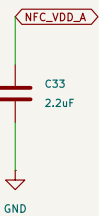
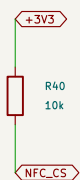
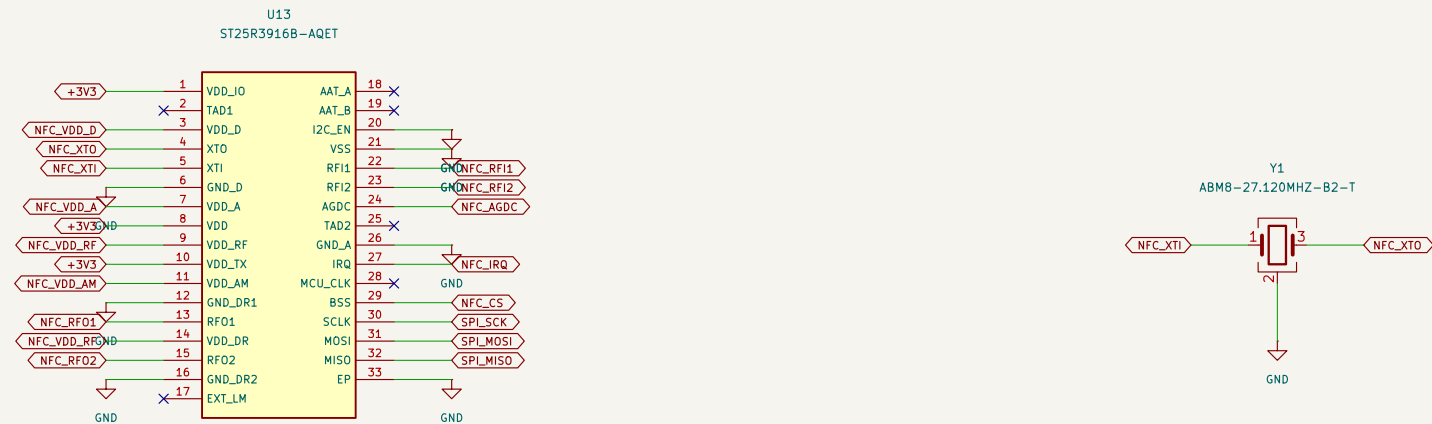
Size: A3  
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft  
Id: 16/25

NFC reader, oscillator and local supplies

ST25R3916B needs a tuned case-mounted loop and TX/RX matching network. Verify crystal load against the selected crystal.



Circuit review in progress – not released for fabrication

Sheet: /NFC reader, oscillator and local supplies/  
File: nfc.kicad\_sch

**Title: NFC reader, oscillator and local supplies**

Size: A3

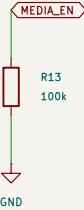
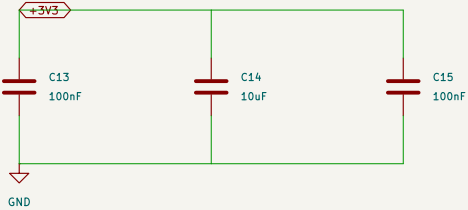
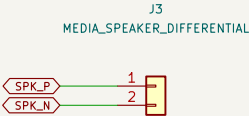
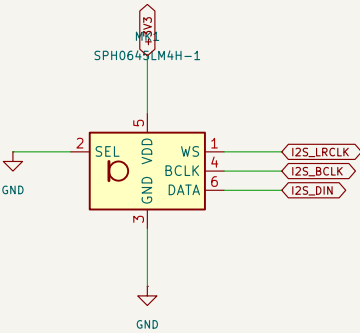
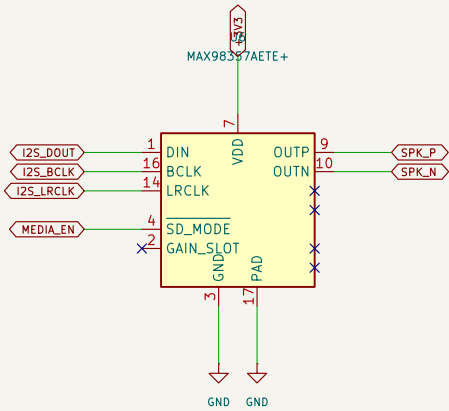
Date: 2026-09-24

Rev: A – engineering draft

Id: 17/25

Media speaker amplifier and digital microphone

Top-mounted media speaker uses differential outputs: neither speaker wire is ground. Digital microphone is case-bottom mounted.



Circuit review in progress – not released for fabrication

Sheet: /Media speaker amplifier and digital microphone/  
File: audio.kicad\_sch

**Title: Media speaker amplifier and digital microphone**

Size: A3 Date: 2026-09-24

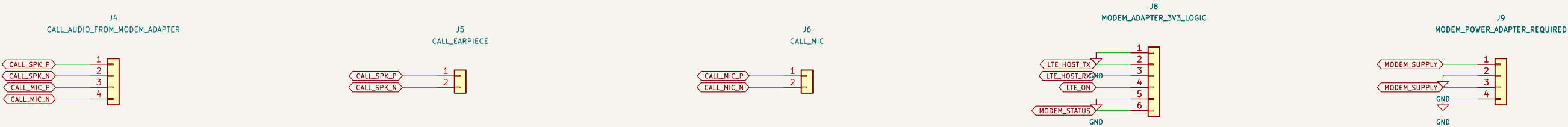
KiCad E.D.A. 10.0.6

Rev: A – engineering draft

Id: 18/25

Cellular adapter and call–audio harnesses

Replaceable modem adapter owns SIM, cellular RF and modem–level translation. No universal–carrier guarantee. Speaker top / mic bottom.



Circuit review in progress – not released for fabrication

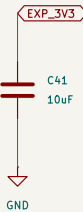
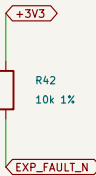
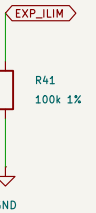
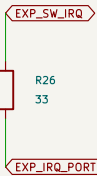
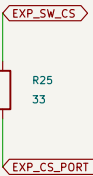
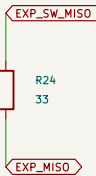
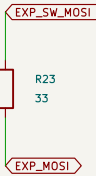
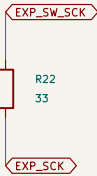
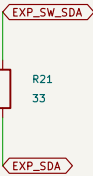
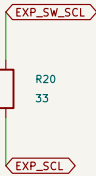
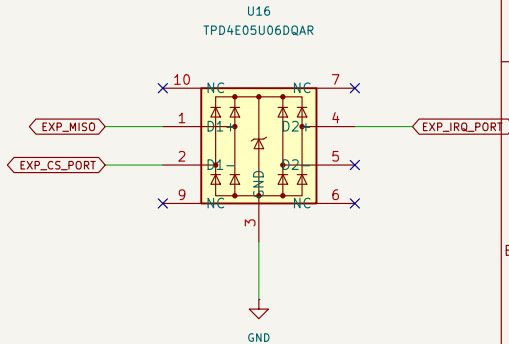
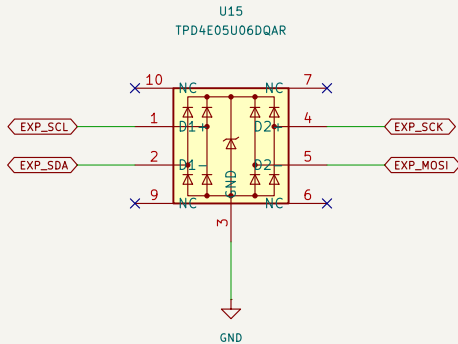
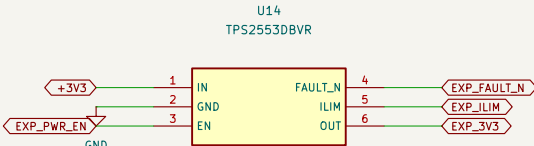
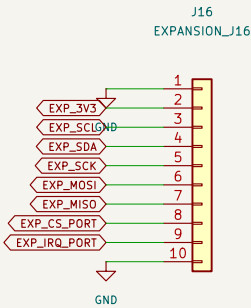
Sheet: /Cellular adapter and call–audio harnesses/  
File: interfaces.kicad\_sch

**Title: Cellular adapter and call–audio harnesses**

|                     |                  |                            |
|---------------------|------------------|----------------------------|
| Size: A3            | Date: 2026–09–24 | Rev: A – engineering draft |
| KiCad E.D.A. 10.0.6 |                  | Id: 19/25                  |

Protected external module port

Port signals pass through U34 on expansion-control. U14 EN defaults low; fault drives U33 and the hardware signal interlock.  
No hot-plug approval.



Circuit review in progress – not released for fabrication

Sheet: /Protected external module port/  
File: protection.kicad\_sch

**Title: Protected external module port**

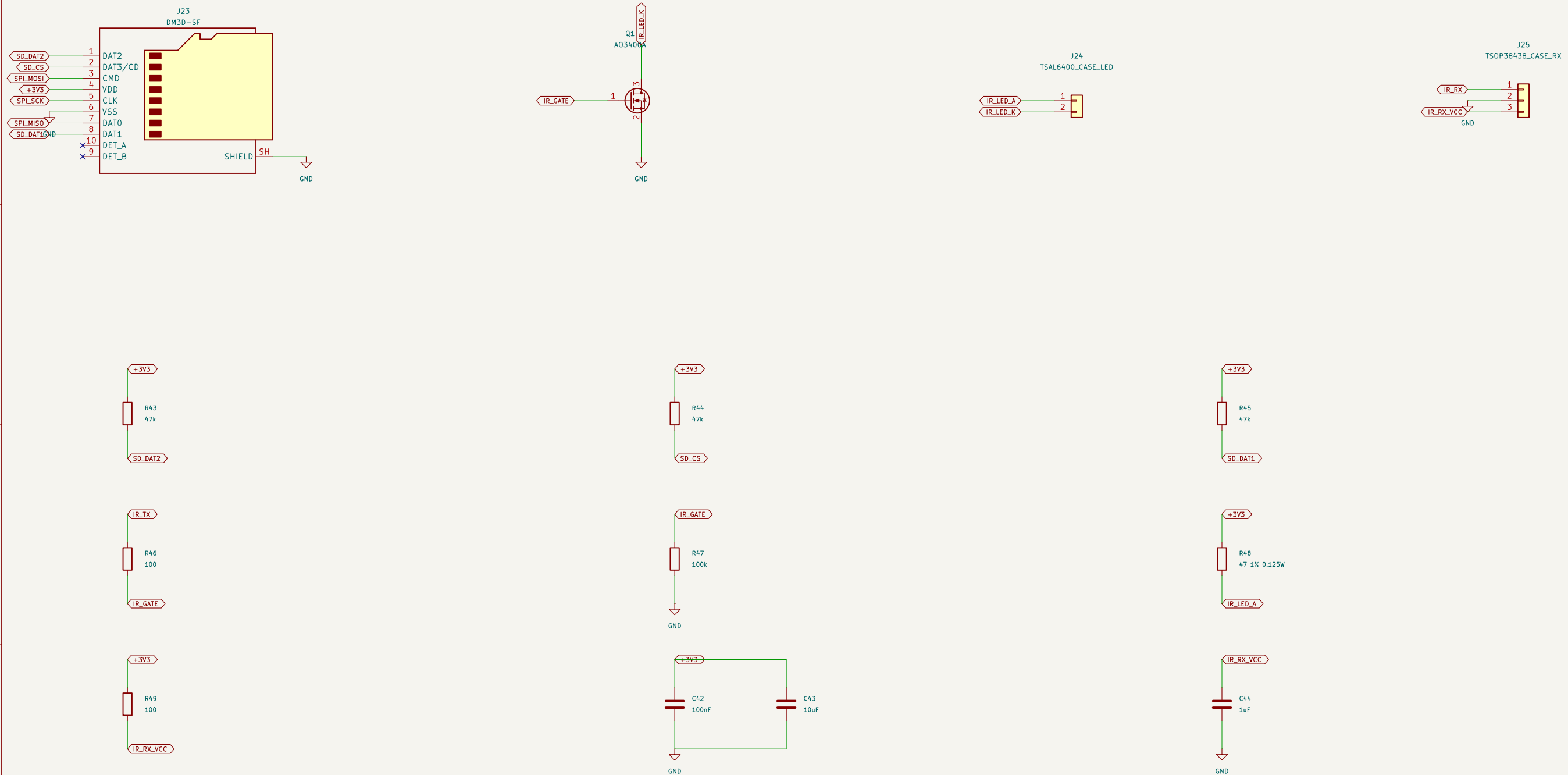
Size: A3  
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft  
Id: 20/25

MicroSD and case-mounted infrared

SPI microSD, transistor-driven IR LED, filtered receiver supply. Confirm case LED/receiver polarity and GPIO3 boot behavior.



Circuit review in progress – not released for fabrication

Sheet: /MicroSD and case-mounted infrared/  
File: storage-ir.kicad\_sch

Title: MicroSD and case-mounted infrared

Size: A3

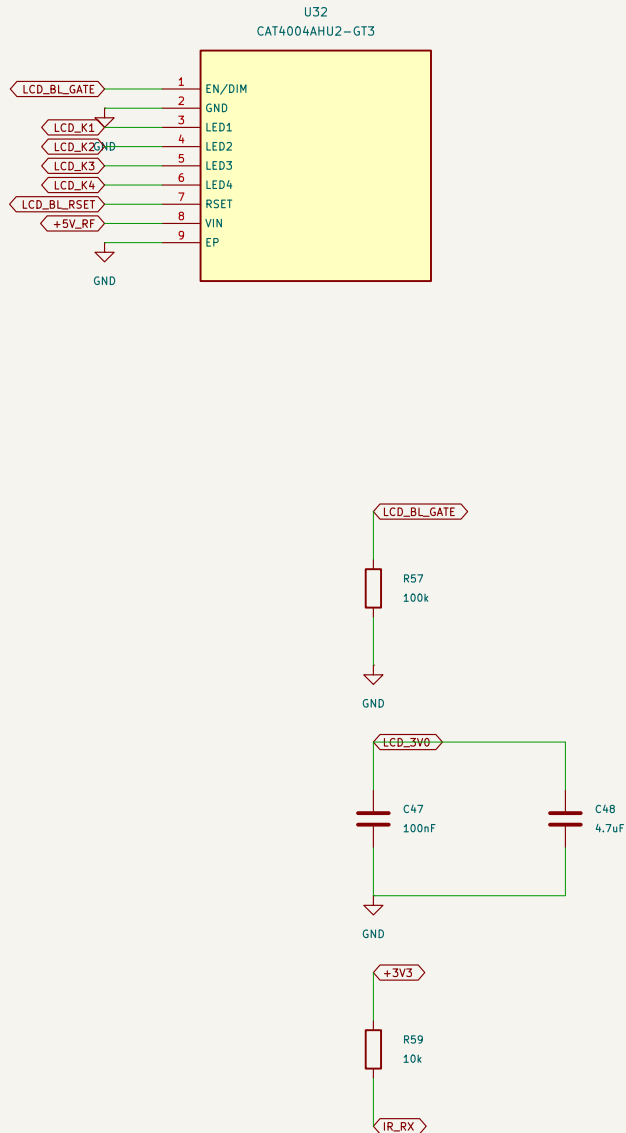
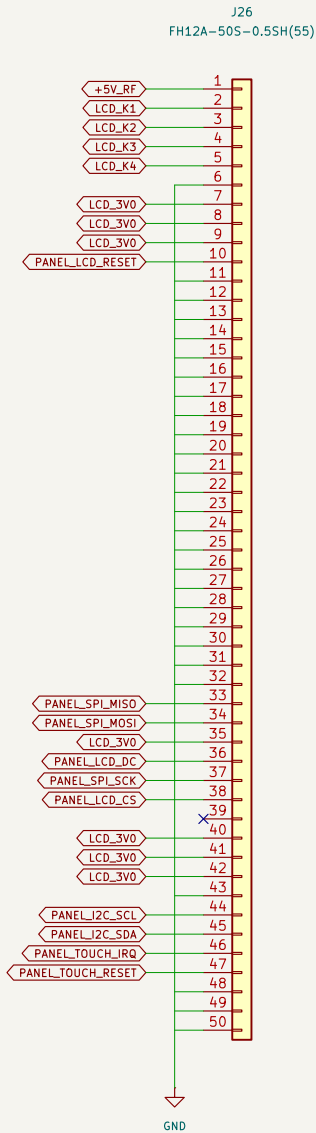
Date: 2026-09-24

Rev: A – engineering draft

Id: 21/25

External display, touch and backlight

CAT4004A: four current sinks, 4.99k sets approx. 15mA/channel. On/off only until dimming timing is qualified. Do not fabricate.



EN/DIM: external R57 pull-down retains default off.  
Initialize U2 GPIO low for >=10ms before enabling.  
Keep high for full-scale current; do not apply ordinary PWM.  
RSET gain, current drift, heat and purchased panel remain unqualified.

Circuit review in progress – not released for fabrication

Sheet: /External display, touch and backlight/  
File: display.kicad\_sch

**Title: External display, touch and backlight**

Size: A3  
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft  
Id: 22/25

Always-on charging supervisor

Separate supervisor selected. U3 remains the charger. Main host powers off at SW19. Circuit capture only; routing, STM32 firmware and fault qualification pending.

A

B

C

D

E

F

A

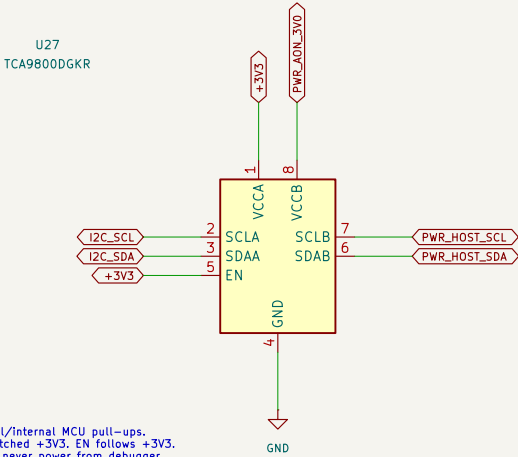
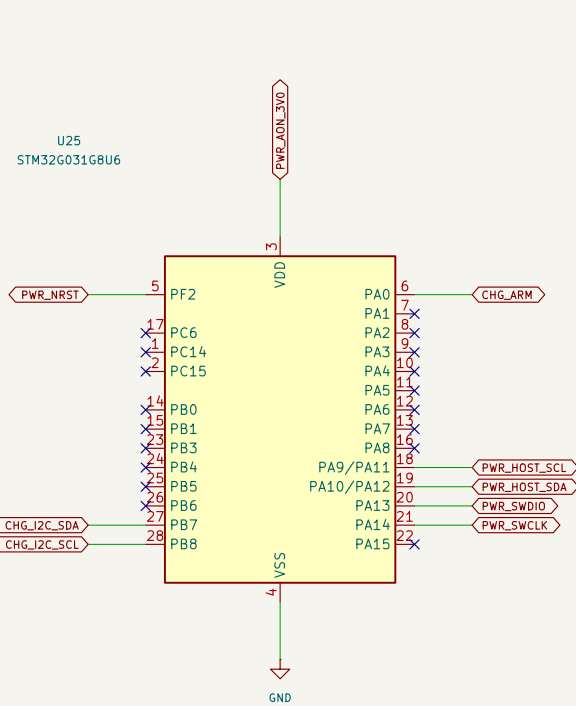
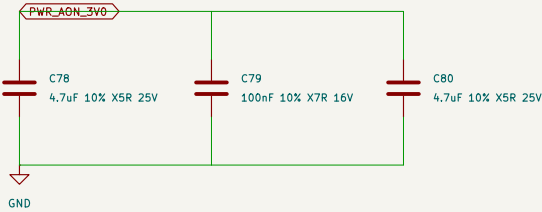
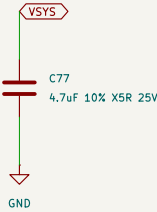
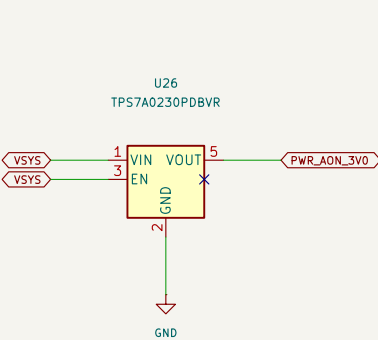
B

C

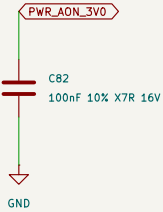
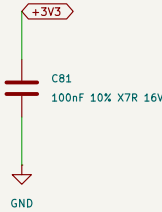
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E

F

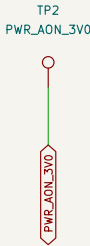
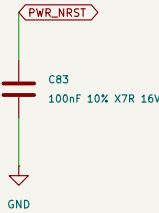
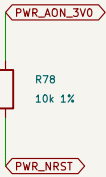
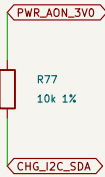
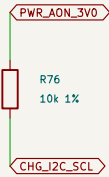


TCA9800 B side: NO external/Internal MCU pull-ups.  
A-side pull-ups are on switched +3V3. EN follows +3V3.  
Debug VTREF is sense only; never power from debugger.



PA0: dedicated active-high charge arm. Reset analog/high-Z allows R79 at Q8 to hold charging disabled independently of U24. Never retain this output through standby or run bootloader while armed.

I2C1: PB8/PB7 AF6 -> charger 0x6A + expander 0x41.  
I2C2: remap PA9/PA10 to PA11/PA12, AF6.  
Host slave address 0x42 reserved; protocol not implemented.



Circuit review in progress – not released for fabrication

Sheet: /Always-on charging supervisor/  
File: power-supervisor.kicad\_sch

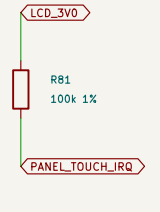
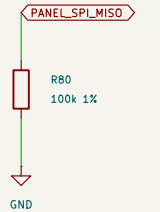
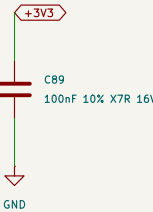
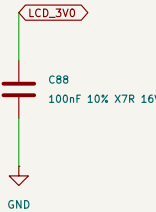
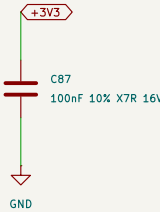
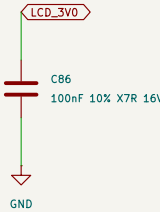
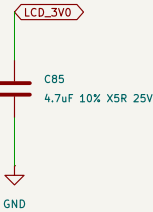
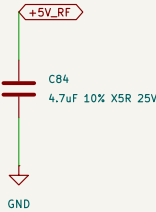
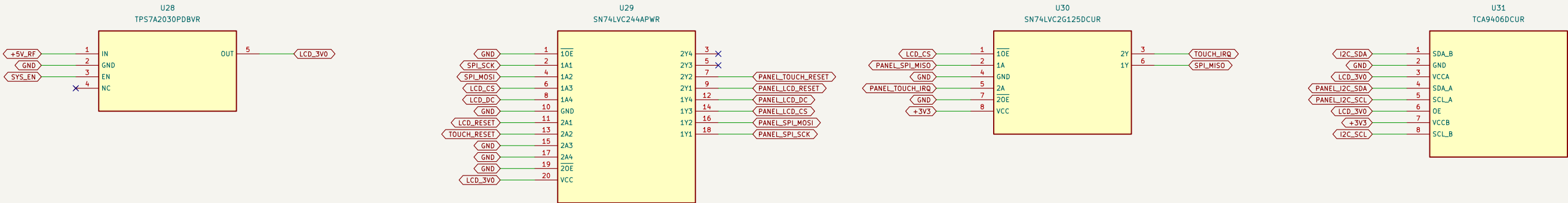
**Title: Always-on charging supervisor**

Size: A3  
Date: 2026-09-24

Rev: A – engineering draft  
Id: 23/25

Display supply and signal translation

Interface routed and continuity checked. Qualify loads, timing, startup and case cable. Do not fabricate.



U29: 3.3 V tolerant inputs; outputs use LCD\_3V0.  
U30: MISO OE follows host LCD\_CS; IRQ buffer always enabled.  
U31: internal pull-ups; verify bus capacitance and touch option.  
U28: 5 V input provides headroom; qualify load and heat.

Circuit review in progress – not released for fabrication

Sheet: /Display supply and translation/  
File: display-interface.kicad\_sch

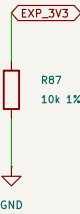
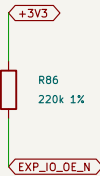
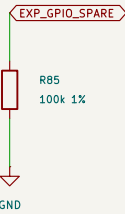
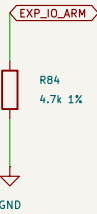
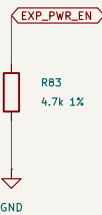
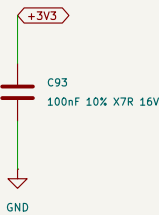
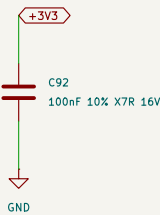
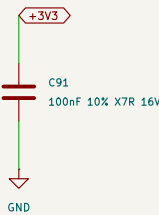
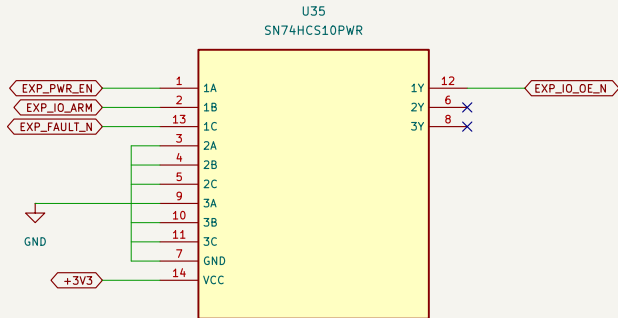
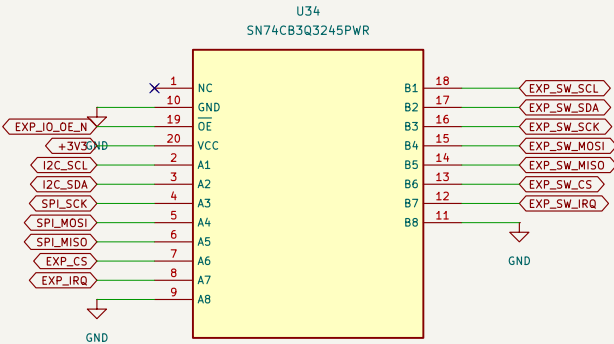
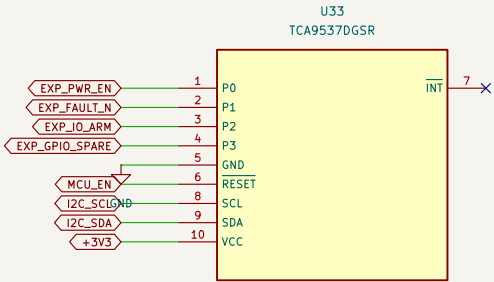
Title: Display supply and signal translation

Size: A3 Date: 2026-09-24  
KiCad E.D.A. 10.0.6

Rev: A – engineering draft  
Id: 24/25

Expansion power control and signal isolation

TCA9537 0x49: P0 power, P1 fault, P2 IO arm, P3 unused input. MCU\_EN low resets all ports to inputs. External pull-downs retain OFF.



U35: /OE = NOT (EXP\_PWR\_EN AND EXP\_IO\_ARM AND EXP\_FAULT\_N).  
Initialize output latch to 0xF0 before direction 0xF1. Power-only 0xF1: armed 0xF5.  
Fault is polled; no MCU GPIO/strap changes. A stuck host I2C bus requires hardware reset.  
No rail-good sensor; qualify load/startup and share buses only after settling with fault clear.  
Accessory address 0x49 is reserved. Swap modules only with handset power off.

Circuit review in progress – not released for fabrication

Sheet: /Expansion control and isolation/  
File: expansion-control.kicad\_sch

**Title: Expansion power control and signal isolation**

|                     |                  |                            |
|---------------------|------------------|----------------------------|
| Size: A3            | Date: 2026-09-24 | Rev: A – engineering draft |
| KiCad E.D.A. 10.0.6 |                  | Id: 25/25                  |